MIL-G-62487(AT)
6 July 1987
SUPERSEDING
(see 6.6)

MILITARY SPECIFICATION

GYROCOMPASS, NORTH SEEKING, FIRE SUPPORT TEAM VEHICLE (FISTV), M981

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the interface, performance, and test requirements for the North Seeking Gyrocompass (NSG) assembly for the Fire Support Team Vehicle (FISTV) system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which My be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 1290

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SPECIFICATIONS
MILITARY

MIL-E-4158	- Electronic Equipment, Ground; General Requirements for.
MIL-E-6051	- Electromagnetic Compatibility Requirements, Systems.
MIL-W-8604	- Welding, Fusion; Aluminum Alloys; Process and Performance of.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, Corrosion and Heat Resistant Alloys; Process for.
MIL-P-11268	- Parts, Materials and Processes Used in Electronic Equipment.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-C-62479	- Control and Display, Targeting Station, M981 (FISTV).

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection of Attributes.
MIL-STD-108	- Basic Requirements for Enclosure for Electrical and Electronic Equipment.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-sTD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Emission and Susceptibility, Test Methods for.
MIL-STD-642	- Identification Marking of U.S. Military Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration System Requirements.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein- Unless otherwise specified; the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

12306660	North Seeking Gyrocompass Assembly.
12306661	Platform and Stop Assembly.
12306662	Housing Assembly.
12306663	Top Cover Assembly.
12306664	Bottom Cover Assembly.
12306670	DC/DC Converter Circuit Card Assembly.
12306675	Power Supply Circuit Card Assembly.
12306680	Analog Servo Electronics Circuit Card Assembly.
12306685	Azimuth Drive Electronics Circuit Card Assembly.
12306690	Resolver/Digital - Converter Circuit Card Assembly.
12306695	Controller/Controller and Interface Circuit Card Assembly.
13030475	Bubble Assembly.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a Part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

Rs-232	-Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interface, August 1969.
RS-422	- Electrical Characteristic of Balanced Voltage Digital Interface Circuits, 1975.

(Applications for copies should be addressed to the Electronic Industries Association, 2001 Eye Street,. NW, Washington, DC 20006.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish NSG units which shall be subjected to first article inspection (see 4.3). First article Inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent NSG units delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. NSG materials shall comply with applicable portions of MIL-E-4158 and MIL-P-11268 (see 4.8.1).

3.2.1 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications, or welding procedures which have been approved by the procuring activity. Welding of aluminum alloys shall be in accordance with MIL-W-8604. Welding of steel shall be in accordance with MIL-W-8611 (see 4.8.1 and 4.8.2).

3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4).

3.3 Design and construction. The NSG shall be fabricated and assembled in accordance with drawings, parts lists and other documents listed on drawing 12306660 (see 4.8.1 and 4.8.2).

3.3.1 Configuration. The NSG assembly (12306660) shall consist Of the following components (see 4.8.1 and 4.8.2):

a. Platform and stop assembly (12306661).

b. Housing assembly (12306662).

c. Top cover assembly (12306663).

d. Bottom cover assembly (12306664).

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e. DC/DC converter circuit card assembly (12306670).

f. power supply circuit card assembly (12306675).

g. Analog servo electronics circuit card assembly (12306680).

h. Azimuth drive electronics circuit card assembly (12306685).

i. Resolver/Digital - digital/analog converter circuit card assembly (R/D - D/A) (12306690).

j. Controller/controller and interface circuit card assembly (12306695).

k. Bubble assembly (13030475).

1. Gyro and buffer amplifier assemblies.

3.3.2 Interface definition. The NSG shall automatically interface with the MIL-C-62479 Targeting Station Control and Display (TSCD). The azimuth and vertical angles shall be displayed, and shall be available for transmittal to other systems as desired (see 4.8.2 and 4.8.3).

3.3.3 Mechanical link. The connector shall be of the M26482 series and shall mate with an MS3476L18-32S cable connector. The pin assignment shall be as shown In Table I. The interconnecting cable is multiple cable, shielded, and twisted pair wire no less than 24 AWG. All external digital lines shall be capable of operating with a nominal cable length of 25 feet (see 4.8.2).

TABLE I. Pin Assignment.

Cable Connector MS3474L18-32P
(mates with MS3476L18-32S)

Pin	Assignment
u	Shield
v	28 VDC Return
w	28 VDC Return
x	+28 VDC Power
Y	+28 VDC Power
z	NSG Status Alert

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TABLE I. Pin Assignment - Continued.

Cable Connector MS3474L18-32P
(mates with MS3476L18-32S)

Pin	Assignment
a	NSG Status Alert
b	TSCD Transmitted Data
c	TSCD Transmitted Data
d	TSCD Received Data
e	TSCD Received Data
f	TSCD Request to Send
g	TSCD Request to Send
h	NSG Clear to Send
j	NSG Clear to Send

3.3.4 Electrical link. The Data Communications Link shall be in accordance with EIA RS-422 standard for a balanced differential interface where signals are transmitted outside of the chassis. The line drivers and receivers shall be in accordance with RS-422, and shall be Advanced Microdevices AM 26LS31 and M 26LS33, respectively, or equivalent circuit. Each signal transmission line shall be terminated at the line receiver with a lK resistor in series with each leg preceded by a termination resistor of 130 ohms as shown in Figure 1 (see 4.8.3).

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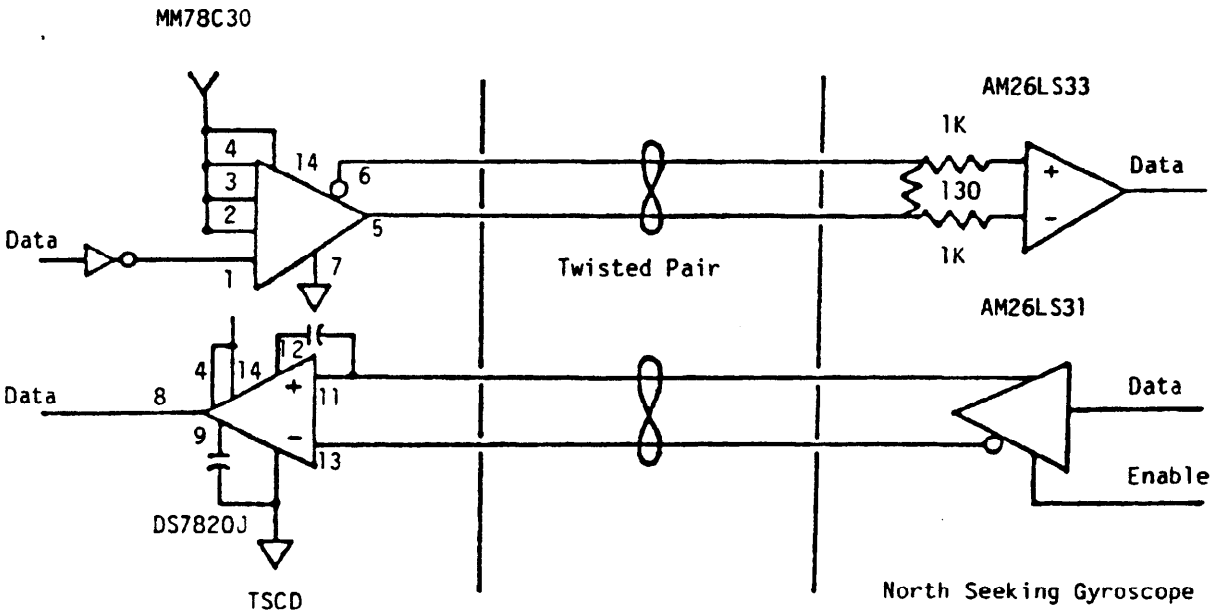


FIGURE 1. Data communications link.

3.3.5 Functional. The interchange circuits shall be in accordance with EIA RS-232, Section 4. Functional Description of Interchange Circuits for the signals are: Transmittal Data, Received Data, Request-to-Send, Clear-to-Send and NSG Status Alert. These signals shall be used to interrogate, accept data from, and handshake with the NSG as shown in Figure 2, Data Communications Definition. The RS-232 binary serial interface need not follow the electrical standard of RS-232 but must be the functional equivalent (see 4.8.3).

3.3.6 Test equipment interface. An NSG test connector shall be provided. Test point designations shall be in accordance with table II (see 4.8.2).

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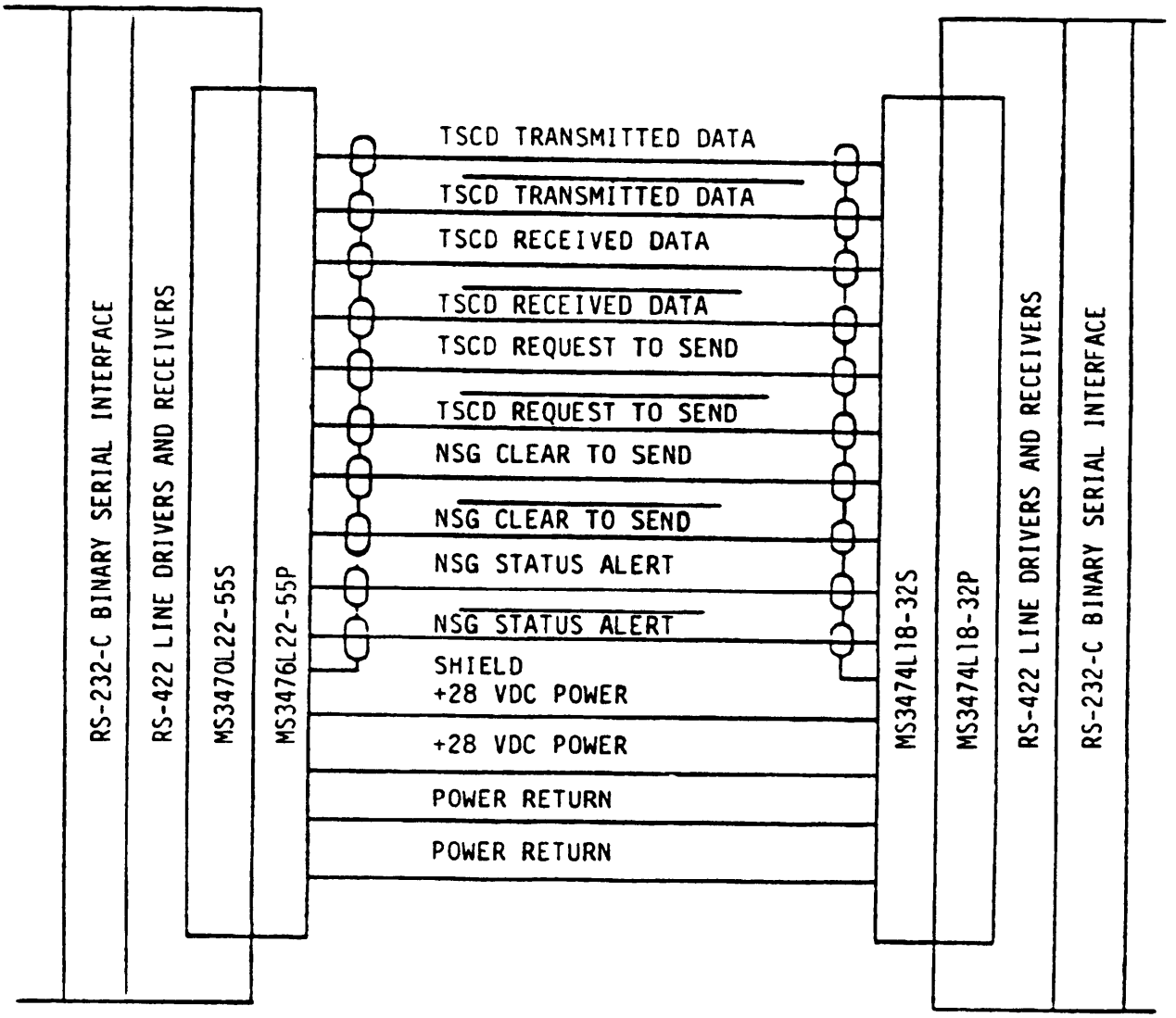


FIGURE 2. Data communications definition.

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TABLE II. Test connector pin designation.

Item	Pin	Designation	Item	Pin	Designation
1	A	AZ Pickoff	32		200 Hz TTL
2	B	EL Pickoff	33	j	100 Hz TTL
3	c	Roll Pickoff	34	k	25 Hz TTL
4	D	Analog Gnd	35	m	2.4 kHz HI
5	E	Resin	36	n	2.4 kHz L0
6	F	+5 VDC	37	P	IATA/NS
7	G	+15 VDc	38	q	IATA/DG
8	H	-15 VDC	39	r	NS
9	J	Mobile Discrete	40	s	153.6 kHz TTL
10	K	Servo Enable Discrete	41	t	Stepper Motor 22V
11	L	EL Gimbal Torquer	42	u	Stepper Motor ØA
12	M	Roll Gimbal Torquer	43	v	Stepper Motor ØA
13	N	+24 VDC	44	w	Stepper Motor ØB
14	P	-24 VDC	45	x	Stepper Motor ØB
15	R	AZ Gimbal Torquer	46	Y	AZ Rate
16	s	-5 VDC	47	z	EL Gyro Torquer
17	T	+12 VDc	48	AA	Roll Gyro Torquer
18	u	200 Hz ØA	49	BB	Az POS/DG Comp
19	v	200 Hz ØB	50	cc	AZ Gyro Torquer
20	w	200 Hz Øc	51	DD	2.4 kHz 6.5V rmS
21	x	800 Hz ØA	52	EE	9.6 kHz 7V rmS
22	Y	800 Hz ØB	53	FF	+10 VDC
23	z	Gyro Temperature	54	GG	-lo VDc
24	a	AZ Pump HI	55	HH	AZ Servo
25	b	AZ Pump LO	56	JJ	Fine Filter
26	c	EL Pump HI	57	KK	Coarse Filter
27	d	EL pump LO	58	LL	EL Servo
28	e	Roll pump HI	59	MM	Roll Servo
29	f	Roll Pump LO	60	NN	EL Erect
30	g	9.6 kHz TTL	61	PP	Roll Erect
31	h	2.4 kHz TTL			

3.3.7 Weight. The weight of the NSG shall not exceed 65 pounds (lb) (see 4.8.4).

3.3.8 Boresight calibration. The NSG inertial unit shall be aligned in azimuth to the precision alignment holes within + 1 mil of perpendicular with respect to a line joining the centers of the precision alignment holes. Elevation shall be aligned with the precision alignment plane to within ± 1 mil (see 4.8.2).

3.3.9 Interchangeability. All parts and assemblies shall be interchangeable to the maximum extent consistent with production techniques and lot size. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7 (see 4.8.2).

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3.4 **Performance.** The performance of the NSG shall be in accordance with the following requirements at standard ambient conditions and the selected operating conditions specified in 3.4.9 and subparagraphs thereof. (Standard ambient condition shall be as defined in MIL-STD-810. except 725 plus or minus 75 millimeters of mercury for atmospheric pressure.) Temporary loss of power during any mode of operation shall not cause physical damage or degradation to the NSG (see 4.8.5).

3.4.1 Preconditioning. Each unit shall be preconditioned to ensure proper workmanship and to exclude infant mortality failures (see 4.8.5.1 through 4.8.5.1.3).

3.4.2 **Power system interface.** The NSG shall not be damaged while operating from a power source functioning at the worst case limits of MIL-STD-1275. Average power consumption shall be less than 120 watts (W). Power shall be no greater than 200 w for more than 5 minutes per hour. The NSG shall function as specified with voltages between 19.5 and 30 volts (V) direct current (de). Whenever the voltage drops below 19.5 V, but not below 12.0 V, and then returns to normal operating voltage, the NSG shall automatically enter the Initialization mode (IM). If the voltage drops below 12.O V dc and returns to normal operating voltage, the operator will have to initiate the reinitialization (see 4.8.5.2 through 4.8.5.2.5).

3.4.3 Data format requirements.

3.4.3.1 Data input. The NSG shall transmit and accept digital data inputs. The data shall be available on a serial digital bus and shall meet the requirements of 3.4.3.1.1 through 3.4.3.1.8.3 (see 4.8.3).

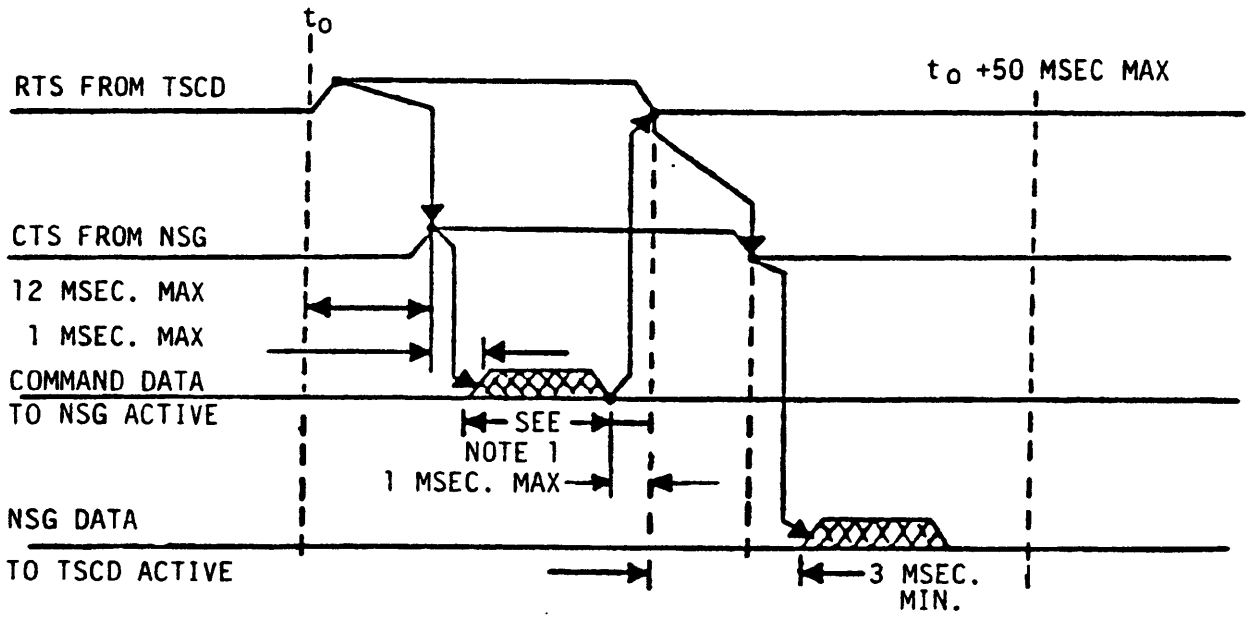
3.4.3.1.1 **Data link control.** The control of data transmitted to and from the NSG on TSCD transmitted and received data lines. respectively, shall be in accordance with EIA RS-232, Sections 4 and 5 for the signals used in Figure 2 for interface Type D, half duplex operation. Data communications shall be made with only one device transmitting at a time. The NSG shall respond to commands or requests but will not independently initiate transmissions. Handshaking shall be accomplished with the TSCD Request-to-Send and NSG Clear-to-Send signals. No path control is required for the signal data link (see 4.8.3).

3.4.3.1.2 Characteristics of information transferred. Data transmitted to and from the NSG on the TSCD transmitted data and received data lines shall use EIA RS-232 conventions as noted below (see 4.8.3):

a. Signal levels shall be identified as mark (1) and space (0).
b. During periods of no transmission, a line shall be held at the marking state.

3.4.3.1.3 **Data rate.** Data transmission rate shall be 19 200 ± 0.1 percent (%) baud (bits per second). Data shall be transmitted in no more than the times specified in Figure 3 (see 4.8.5.3).

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THIS CHART SHOWS TIME SEQUENCING AND
NOT NECESSARILY VOLTAGE LEVELS.

NOTE 1; MESSAGE LENGTHS ARE AS FOLLOWS

P	1.5 MSEC MAX
I	1.5 MSEC MAX
R	1.5 MSEC MAX
X	3.0 MSEC MAX
U	9.0 MSEC MAX

FIGURE 3. Message/Response sequence.

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3.4.3.1.4 **Character format.** All character transmission shall be ASCII encoded as specified in table 111 (see 4.8.3).

TABLE III. 64-Character ASCII.

RS-232 Character Code and Equivalent Chart

Character	7-bit ASCII		7-bit ASCII	Character	7 bit ASCII	Character	7-bit ASCII
@	40	P	50	Space	20	0	30
A	41	Q	51	!	21	1	31
B	42	R	52	"	22	2	32
C	43	S	53	#	23	3	33
D	44	T	54	$	24	4	34
E	45	u	55	z	25	5	35
F	46	v	56	&	26	6	36
G	47	w	57		27	7	37
H	48	x	58	;	28	8	38
I	49	Y	59	)	29	9	39
J	4A	z	5A	*	2A	.	3A
K	4B	[	5B	+	2B	;	3B
L	4C	\	5C		2C	<	3C
M	4D	1	5D		2D	-	3D
N	4E	4	SE		2E	>	3E
o	4F	←	SF	./	2F	?	3F

3,4.3.l.4.l **Data character.** Data shall be transmitted serially by character. Each character is a 10-bit word. Each 10-bit word shall consist of a start bit (space), a 7-bit ASCII encoded character (transmitted LSB first), a parity bit, and a stop bit (mark). Odd parity shall be used (see 4.8. 3).

3.4.3.1.4.2 **Numerical data format.** All numeric data will be sent MSB first (see 4.8.3).

3.4.3.1.4.3 **Numerical encoding.** Unless otherwise specified herein, all numeric data will be transmitted as ASCII encoded decimal digits (see 4.8.3).

3.4.3.1.4.4 **Update format (U).** The NSG shall accept, respond to, and utilize update digital data transmitted by the TSCD having the following format:

eeeeeesnnnnnnn

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where:

a. eeeeee represents Easting data.

b. s represents the sign. An ASCII "1" shall indicate a negative number and an ASCII "O" shall indicate a positive number.

c. nnnnnnn represents Northing data.

Utilization and response to Update Data shall be described in table IV and elsewhere herein. Update Data shall consist of Universal Transverse Mercator (UTM) vehicle position coordinates. Northings south of the equator will be transmitted as actual UTM map coordinates preceded by an ASCII numeral "1". Leading zeros will be transmitted. Resolution of data will be 1 meter (see 4.8.5.4).

TABLE IV. Messages.

Type of message	From TSCD to NSG		From NSG to TSCD			See note
			NSG Mode			
		S/00	I 01, 02	R 04	D 03	2
Initialize	I<	>Y<	>Y<	>Y<	>Y<	
Realign	R<	>N<	>N<	>N<	>Y<	
Update	Ueeeeeesnnnnnnn	>Y<	>Y<	>Y<	>Y<	
Request for Polar Examine	P<	>N<	>N<	>N<	>aaaawvv<	
XhhB<	>mm.. .m<	>mm. .● m<	>mm. ..m<	>mm...m<		1

Notes:

1) "hh" is a pair of encoded hexadecimal digits specifying a starting NSG RAM location, "B" is an ASCII encoded hexadecimal digit specifying the number of memory locations to be read, and "m...m" is the contents of the specified memory locations in ASCII encoded hexadecimal pairs of digits. See table V.

2) Mode number shall be as specified in figure 4.

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TABLE V. Examine messages.

Message purpose	From TSCD to NSG	From NSG to TSCD	See note
Data Link Test	X027 <	> eeeeeesnnnnnnn <	1
Status Request	XO02 <	> ccbb <	2
Examine Memory	XhhB <	> mm...m <	
Examine Status	Xoll <	> C c <	

Notes:

1. Eastings and Northing are being utilized.

2. cc defines mode number (Figure 4): bb defines failure number (see table VII) .

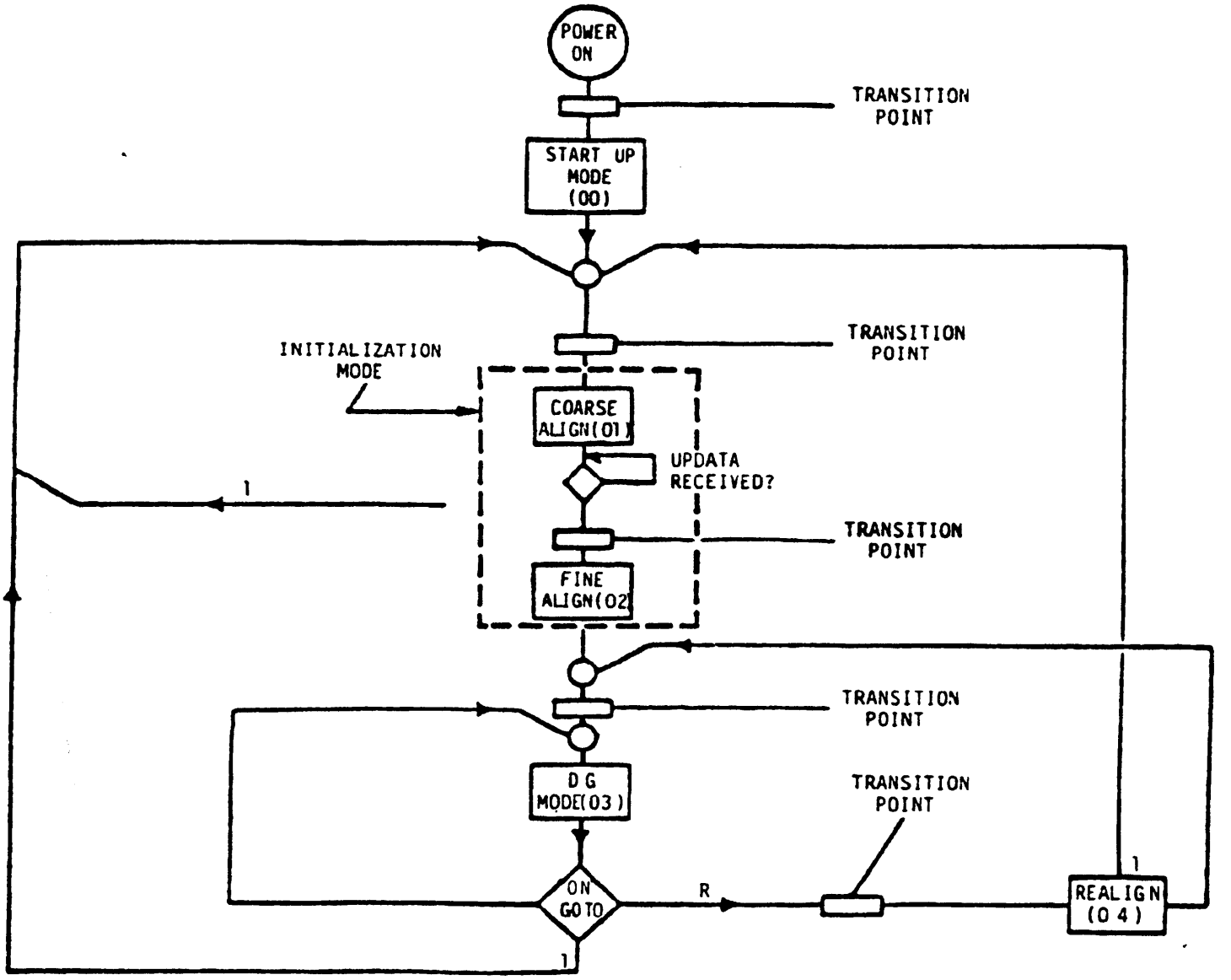
3.40301.5 Timing. Transmissions shall be accomplished in accordance with Figure 3. Upon completion of any message/response sequence, the TSCD shall be capable of participating in another message/response sequence except as noted below (see 4.8.5.3):

a. During Start Up, Initialize, and Realign modes, the average frequency of message/response sequences shall not exceed one per 10 seconds.

b. A new sequence shall not be started until a current sequence is completed.

3.4.3.1.6 Data range. The range of expected data is delineated in table VI (see 4.8.3).

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NOTE 1; MODE NUMBERS ARE SHOWN IN PARENTHESIS.

FIGURE 4. NSG Modes.

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TABLE VI. North Seeking Gyroscope

Description	Numerical value range (decimal)
Easting	120,000 - 880,000m
Northing	O-9,999,999m
Target grid azimuth	O-6399 roils
Target vertical angle	0-1209 roils 6399-5457 milS
Hemisphere of operation	In north hemisphere, UTM map northings preceded by ASCII "O". In south hemisphere, UTM map northings preceded by ASCII "1".

3.4.3.1.7 TSCD to NSG. The NSG shall accept, respond to, and utilize transmissions from the TSCD having the following format whenever operating power has been applied for more than 1 second (see 4.8.3).

(MESSAGE TYPE) (OPTIONAL DATA) (END DELIMITER)

3.4.3.1.7.1 Message type. The following message types are legitimate (see 4.8.5.5):

a. I - commands NSG to initialize
b. R- commands NSG to realign
co u - commands NSG to update true north grid north conversion
d. P- commands NSG to provide polar coordinate data
e. x - commands NSG to provide contents of specified memory location

Transmission of other message types shall be treated as an error.

3.4.3.1.7.2 Optional data. Optional data shall be transmitted only with U and X message types. Transmission of optional data with other message types shall be treated as an error (see 4.8.5.6).

3.4.3.1.7.3 End delimiter. The end delimiter shall be ASCII < (less than). Transmission of any other delimiter shall be treated as an error (see 4.8.5.6).

3.4.3.1. NSG to TSCD. The NSG shall respond with transmissions having the following format (see 4.8.3):

(START DELIMITER) (DATA) (END DELIMITER)

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3.4.3.1.8.1 Start delimiter. The start delimiter shall be ASCII > (greater than). Transmission of any other delimiter shall be treated as an error (see 4.8.3).

3.4.3.1.8.2 Data. The NSG shall respond with the following data transmissions (see 4.8.3):

a. Y - indicates transmission received and execution shall be as specified.
b. N- indicates transmission received but transmission is a control error and shall not be executed due to MODE incompatibility.
co ? - indicates error in transmission, see paragraph 3.4.7.1.
d. aaaavvvv - Polar data; see paragraph 3.4.5.1.
e. mm...m - ASCII encoded hexadecimal pairs of digits; see Examine, paragraph 3.4.5.2.

3.4.3.1.8.3 End delimiter. The end delimiter shall be ASCII < (less than). Transmission of any other delimiter shall be treated as an error (see 4.8.3).

3.4.4 NSG modes of operation. The NSG shall react to stimuli from the TSCD as provided in this document and as described in table IV (see 4.8.3).

3.4.4.1 NSG modes. The NSG shall exhibit four modes as followed and as shown in Figure 4 (see 4.8.5.5).

3.4.4.1.1 Start Up mode. Upon receipt of operating power, the NSG shall enter into a turn on sequence in which power supply transients are allowed to settle, circuit hardware/software initial values are set, leveling and spin up as required are accomplished, and all preparations are made in anticipation of entering into Initialization Mode (IM). The Start Up mode shall be completed in less than 3 minutes; the Start Up time is designated T_1. The NSG shall be capable of accepting data transmissions and returning responses within 1 second of application of operating power. Upon completion of the Start Up function, the NSG shall proceed directly to the IM. In the event that a realign command or request for polar data occurs while in the Start Up mode, the NSG shall respond with >N< indicating a control error (see 4.8.5.7).

3.4.4.2 IM. The IM shall incorporate two submodes: Coarse Align and Fine Align. The time allowed for Coarse Align is designated T_2. The time allowed for Fine Align is designated T_3 (see 4.8.3).

3.4.4.2.1 Timing As shown in figure 5A, whenever an update has been transmitted anytime prior to entering the Fine Align Submode, the NSG shall not wait and shall proceed. The sum of $T_1 + T_2 + T_3$ shall not exceed 10 minutes when entering IM from Start Up; the sum of $T_2 + T_3$ shall not exceed 10 minutes when entering IM from any mode other than Start Up (see 4.8.5.7).

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3.4.4.2.2 Waiting As shown in Figure 5B, In the event that no update has been received, the NSG shall wait for an update transmission prior to proceeding with the Fine Align submode. The wait time is designated T_w.

When an update command has been received, the NSG shall proceed. Timing requirements are the same as defined in paragraph 3.4.4.2.1 (see 4.8.5.8).

3.4.4.2.3 Polar data. Polar data responses shall be generated using the most recent update data. The NSG shall accept data and return responses while in IM. During IM, the NSG shall acquire true North and vertical and accomplish all necessary actions and preparations required to respond as specified to requests for polar data. Upon completion of IM, the NSG shall automatically enter into the Direction-Gyro Mode (D-GM). Receipt of an Initialization Command shall cause initialization to start or restart and proceed independent of the mode then in effect including IM but excluding Start Up. While in IM the NSG shall not proceed to any mode other than IM or D-GM nor shall the NSG transmit invalid data. Therefore, while in IM, the NSG shall respond to commands for Realign and requests for Polar Data with >N< indicating a control error (see 4.8.5.9).

3.4.4.3 D-GM. After the NSG has been initialized it shall, upon command, provide the azimuth and vertical angle of the targeting stations while the targeting station is in motion or stationary. The targeting station will be used to track moving targets, which can have a crossing velocity as high as 30 milliradians per second (mrad/see). The NSG shall provide "real time" measure of the angular position of these targets. . Any transport lag in the availability of this angular data shall be considered as part of the basic accuracy of this instrument. While in D-GM, the NSG shall be capable of going to any other mode in response to legitimate stimuli from the TSCD. The NSG shall accept data and return responses while in D-GM (see 4.8.5.10).

3.4.4.4 Realign mode (RM). Upon receipt of a Realign command, the NSG shall enter into a RM. Realignment shall be accomplished within 5 minutes of the transmission of the realign command from the TSCD. The NSG shall only accept a realign command when in the D-GM. If a realign command is received by the NSG when it is in a mode other than D-G, the NSG shall transmit >N< indicating a control error. While in RM, the NSG shall accept data and return reponses. While in RM, the NSG shall not transmit invalid data; therefore, the NSG shall respond to requests for polar data with >N< indicating • control error. During RM, the NSG shall reduce accumulated bias drift and/or apparent drift due to vehicle coordinate changes to within specified accuracy limits when such drift is less than 30 mrad at the time RM is commanded. Upon completion of the RM sequence, the NSG shall automatically enter into D-GM. The NSG shall be capable of entering IM while in RM in response to an initialization command (see 4.8.5.6).

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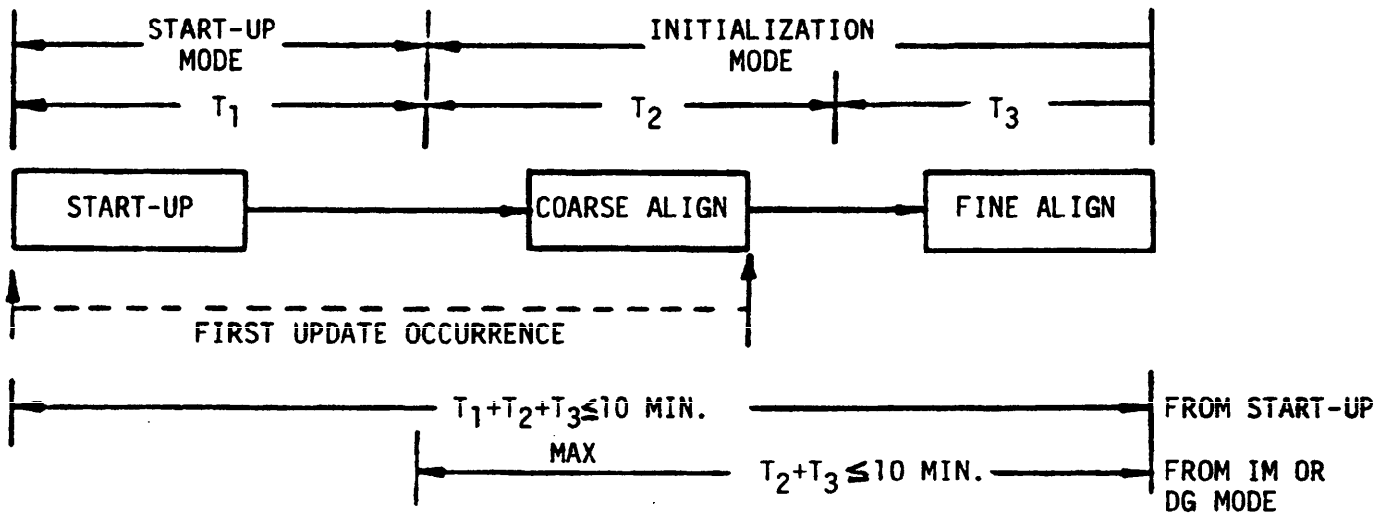


FIGURE 5A

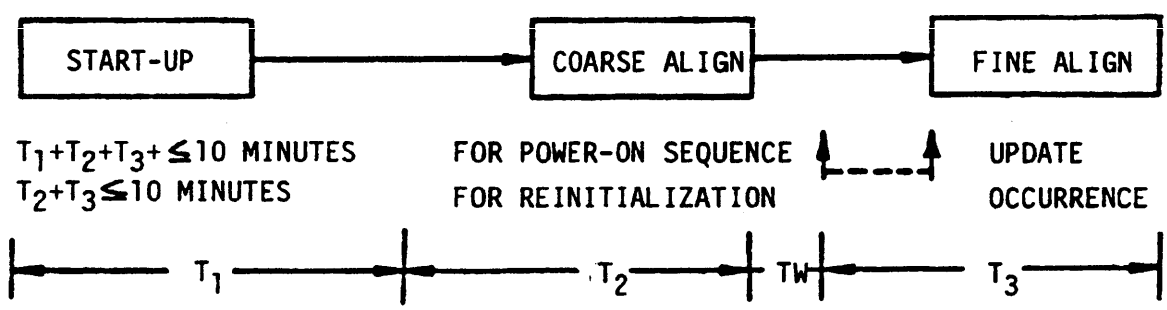


FIGURE 5B

FIGURE 5. Initialization mode timing.

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3.4.5 Data request. The NSG shall respond to data requests as follows:

3.4.5.1 Polar data (P). The NSG shall respond to TSCD requests for Polar Data. Responses from the NSG to Polar Data requests shall be in accordance with table IV. Polar data shall be in roils (6400 roils per 360°) and shall be corrected for and referenced to UTM grid north for the most recently TSCD transmitted update data (vehicle grid location). Grid north shall be O roils and increase for clockwise rotation. Vertical O shall be zero elevation angle and shall increase for positive rotation about pitch. Polar data shall not be expressed as a negative value. Leading zeroes shall be transmitted. Polar data shall be transmitted as ASCII encoded decimal characters in a block:

aaaavvvv

four azimuth characters followed by four vertical characters, The resolution of polar data shall be 1 roil. The NSG shall sample, compute, and transmit polar data in accordance with the timing in Figure 3 (see 4.8.5.3).

3.4.5.2 Examine (X). The NSG shall respond to EXAMINE requests in accordance with table IV and table V (see 4.8.5.4).

3.4.6 Accuracy. The NSG, while in the D-GM, shall provide the following accuracies for any combination of system parameters and requirements as specified herein (see 4.8.5.10 and 4.8.5.10.1).

3.4.6.1 Azimuth error. The azimuth error shall not exceed 8.5 roils . (RMS) after 1.0 hour in the dynamic mode of operation and at latitudes up to 75 degrees. *This* accuracy can be achieved with any realistic combination of initial gyrocompassing accuracy and directional gyro drift or by the use of any other available techniques (see 4.8.5.10 and 4.8.5.10.1).

3.4.6.2 Vertical angle error. The vertical angle error shall not exceed 3.5 roils (RMS) after 1.0 hour operation in the dynamic mode of operation (see 4.8.5.10 and 4.8.5.10.1).

3.4.7 NSG built-in test performance.

3.4.7.1 Error handling. The NSG shall perform sufficient checks on transmission from the TSCD and shall transmit:

>?<

if parity or transmission character count is incorrect, the transmission does not match specified format; or, in general, for any undefined transmission (see 4.8.5.11).

3.4.7.2 Status alert. Whenever a mode change or a change in BIT status occurs, the NSG will set the Status Alert Line to the HIGH state. When mode/bit status request XO02 is received, the NSG shall reset the Status Alert Line. Mode changes are identified as transition points in Figure 4 (see 4.8.5.12).

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3.4.7.3 Status retention. A failure shall not be indicated unless a failure has been encountered. Whenever a failure is encountered, the last failure shall be stored by type and shall not be cleared until a response to XO02 has been transmitted to the TSCD as specified in table V. Mode indication will be contained within the first two hex digits (mm) and bit failure number shall be contained in the second two hex digits (bb) (see 4.8.5.13).

3.4.7.4 NSG internal failures. The failure types listed in table VII shall be detected and stored by type number. Whenever XO02 is transmitted from the TSCD, the NSG shall respond with an ASCII encoded hexadecimal representation of the failure status as per table VII and table V (see 4.8.5.13).

TABLE VII- Failure status.

Hexadecimal type number	Failure type/nomenclature
00 1/	Pass
01	Reserved
02	Reserved
03	Reserved
04	Reserved
05	Reserved
06	Reserved
07	Reserved
08	Excessive D-G to North Seeking Transfer Time
09	Excessive North Seeking to D-G Transfer Time
OA	Gyro Compass Fine Filter Saturation
OB	Reserved
Oc	ROM Sum Check
OD	A/D-D/A Check
OE	Servo Loop Saturation
OF	Excessive Azimuth Gyro Temperature
10	Elevation Range
11	ITA Switch
12	Ram Read/Write
13	Cal Prom Sum Check
14	Temperature Saturation

1/ 00 Hex indicates Pass. Any other indicates a failure has been encountered.

3.4.7.5 Azimuth resolver linearity. With the system in the D-GM ● ode the system shall provide polar data within a one (1) sigma accuracy of + 2 roils as the unit is rotated 360° in azimuth in 15° increments (see 4.8.5.14).

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3.4.7.6 Elevation resolver linearity. With the system in the D-GM mode the system shall provide polar data within a one (1) sigma accuracy of ± 3 roils as the unit is rotated in elevation between +60° and -45° in 15° increments (see 4.8.5.15).

3.4.7.7 Vehicle motion. The NSG shall perform as specified herein when operated within the following vehicle motion envelopes (see 4.8.2).

a. Horizontal velocity (ground speed) - 0 to 40 nautical miles per hour.

b. Vertical velocity - 0 to 15 nautical miles per hour.

c. Acceleration (excluding shock and vibration) - 0 to 1.5g in any direction due to vehicle starting, stopping or turning.

d. Pitch (longitudinal vehicle axis) - -30° to +30° with respect to the horizontal.

e. Roll (traverse vehicle axis) - -20° to +20° with respect to the horizontal.

f. Angular velocities - pitch, roll and azimuth rates - 0-200 deg/sec.

g. Angular accelerations - pitch, roll and azimuth - 6000 deg/sec/sec (max).

3.4.8 Targeting head motion. The NSG shall perform as specified herein when operated within the following launcher motion envelopes (see 4.8.5.16.1 and 4.8.5.16.2):

a. Angular velocities - pitch, roll and azimuth rates - 0-200 deg/sec.

b. Angular accelerations - pitch, roll and azimuth - 600 deg/sec/sec (max).

Cm Launcher head vertical angle - -23° to +38° with respect to the vehicle.

d. Launcher head azimuth angle - 0° to 360" multi-turn with respect to the vehicle.

e. Launcher head stow position - -5° vertical angle; 180° azimuth angle.

f. The launcher head will be stowed at all vehicle velocities greater than 5 mph.

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3.4.9 **Environmental conditions.**

3.4.9.1 Temperature. The NSG shall sustain exposure, in the operating and nonoperating modes, to the following respective temperature ranges without degradation of performance (see 4.8.5.17.1).

a. Nonoperating temperature: +160° to -40°F

b. Operating temperature: +145° to -25°F

3.4.9.2 Humidity. The NSG shall operate in humidity conditions of 95% relative humidity at temperatures from minus 25°F to plus 145°F, and at humidity conditions of 5% relative humidity at temperatures from +70°F to +145°F (see 4.8.5.17.2).

3.4.9.3 Sand and dust. The NSG shall be operational while exposed to wind blown sand particles 0.01mm to 1.00mm in diameter with predominant diameters between 0.15mm and 0.3mm and winds greater than 30 knots; and to wind blown dust with a density of 6×10^{-9} grams/cubic centimeter (g/cm^3) 0.0001mm to 0.01mm in diameter blowing at 35 knots (see 4.8.5.17.3).

3.4.9.4 Salt fog. The NSG shall operate when exposed to sea salt fog fallout of 25 pounds per acre per year (lbs/acre/year) (see 4.8.5.17.4).

3.4.9.5 Water spray. The NSG electrical and electronic equipment shall be spray tight as defined in MIL-STD-108 so that when the vehicle is cleaned by hosing, the performance of the NSG is not degraded. As an alternate, the equipment may be tested in accordance with Water Immersion Test, Method 512, Procedure 1 of MIL-STD-810 (see 4.8.5.17.5).

3.4.9.6 Explosive atmosphere. In operation, the NSG shall not ignite an ambient-explosive-gaseous mixture of air (see 4.8.5.17.6).

3.4.9.7 Electrostatic discharges. The NSG shall be designed to withstand discharge of personnel-borne electrostatic charges of 500 picofarad capacity, at 30 000 volts, discharged through 5000 ohms (see 4.8.5.17.7)*

3.4.9.8 Low pressure altitude. The NSG shall operate at altitudes from 0 to 10 000 feet [14.7 pounds per square inch absolute (psia) to 10.1 psia]. Non-operating, the NSG shall withstand exposure to altitudes from 0 to 40 000 feet (14.7 psia to 2.47 psia) (see 4.8.5.17.8).

3.4.9.9 Temperature shock. The NSG shall be operational after exposure to temperature extremes of -40°F to +160°F (see 4.8.5.17.9).

3.4.9.10 Fungus. The NSG shall be operational after exposure to the fungi specified in MIL-STD-810, method 508 for 90 days at a relative humidity of 95% and a temperature of 84°F (see 4.8.5.17.10).

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3.4.9.11 Rain. The NSG shall be operational after exposure to a 12 hour 9.50 inch rainfall which includes intermittent winds of 35 knots and the following intensities (see 4.8.5.17.11):

Duration	Rainfall Rate
1 minute	27 inches/hour
5 minutes	12 inches/hour
10 minutes	9 inches/hour
1 hour	5.5 inches/hour

3.4.9.12 Snow. The NSG shall be operational during snowfall with a crystal size range of 0.05mm to 20mm diameter, with the median diameter between 2.Omm and 5.Omm and a temperature of 32 ± 5°F and in the light winds. When blown by strong winds (greater than 10 knots), the crystal size range shall be from 0.02mm to 0.4mm diameter (see 4.8.5.17.12).

3.4.9.13 Icing. The NSG shall be operational during and after exposure to the following types of icing (see 4.8.5.17.13). (The NSG shall not be damaged by physical clearing of the ice.)

a. Frost with a surface sublimated with a deposit of ice crystals averaging 1/16 inch to 1/4 inch diameter.

b. Rime ice granules averaging 0.056mm diameter.

c. Clear glaze ice with a 2.0 inch accumulation with a specific gravity of 0.85.

3.4.9.14 Solar radiation. The NSG shall operate when exposed for 4 hours to solar radiation of 360 British thermal units per square foot per hour (BTU/ft^2/hour) with an ambient temperature of 145°F and a wind speed less than 4 knots (see 4.8.5.17.14).

3.4.9.15 Shock and vibration. The NSG, when mounted in the M981 targeting station, shall perform as specified under the shock and vibration associated with all phases of the FISTV'S mobility, including cross-country travel, highway travel, and swimming operations. The NSG shall also survive, with necessary preparation and packing, the shock and vibration associated with transport per paragraph 5.1. The NSG shall be designed to perform as specified after exposure to the following shock and vibration levels.

3.4.9.15.1 Shock levels. Shock levels shall be determined in accordance with MIL-STD-810, method 516, procedure I for ground equipment and either Figure 516 (11 millisecond sawtooth pulse with 40g peak) or Figure 516 (11 millisecond half-sine wave pulse with 30g peak) may be used with the sawtooth shock preferred (see 4.8.5.17.15).

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3.4.9.15.2 Vibration levels. Vibration levels shall be per MIL-STD-810, method 514, procedure VIII, curve W of Figure 514.2-6, and with a minimum sinusoidal cycling time of 30 minutes per 1000 miles. The NSG shall be subjected to a minimum cycling time of 2.5 hours and a maximum of 3 hours. Sweep time shall be 15 minutes from 5 to 5000 Hz. Curve W imposes 1.0 inch double amplitude vibration for 5 to 5.5 Hz; 1.5g from 5.5 to 30 Hz; 0.033 inch double amplitude from 30 to 50 Hz; and 4.2g from 50 to 500 Hz. Equipment not capable of surviving the levels when hard-mounted shall be mounted on absorbers or isolators sufficient to ensure survivability (see 4.8.5.17.16).

3.4.9.16 Electromagnetic interference (EMI/EMC). Compatibility with communication and electronic equipment subsystems and the vehicle system shall be assured consistent with the requirements of this specification and the design of the NSG.

3.4.9.16.1 System Level EMI/EMC. Electromagnetic compatibility shall be controlled in accordance with tactical vehicle provisions of MIL-STD-461. Notice 4 for Class III-A, characteristics RE05 and CE07, and MIL-E-6051 (see 4.8.5.17.17).

3.4.9.16.2 Subsystem Level EMI/EMC. Unless otherwise specified (see 6.2), the NSG shall operate as specified when tested in accordance with the following MIL-STD-462 procedures to the limits given in MIL-STD-461, Notice 4 (see 4.8.5.17.18).

CEO1	Conducted Emission, Power Leads
CE04	Conducted Emission, Power Leads 50 kHz - 76 MHz
Csol	Conducted Susceptibility, Power Leads
CS02	Conducted Susceptibility, Power Leads
CS06	Spike, Power Supply Leads
RE02. 1	Broadband Radiated Emissions, Electric Field 2 MHz - 1 GHz
RSO1	Radiated Susceptibility, Magnetic Field
RS03	Radiated Susceptibility, Electric Field 2 MHz - 10 GHz

3.4.10 Reliability. Manufacturing and production techniques employed in the fabrication of the NSG shall not degrade the inherent reliability. Preconditioning and screening of the end item shall be employed as a means to detect anomalies induced in the fabrication process which would degrade the inherent reliability. The NSG shall have a Mean-Time-Between-Failure (MTBF) of not less than 2000 hours. This is the aggregate demonstrated value for a mature unit (see 4.8.18).

3.4.11 Maintainability. The NSG shall have a Maintenance Ratio (MR) of .0013 for all maintenance actions performed at organizational, direct support (DS), and general support (GS) levels. MR is calculated by dividing the calendar maintenance time by the total system operating time (see 4.8.19).

3.5 Protective finishes. Unless otherwise specified (see 6.2), the exterior surface of the NSG shall be primed per MIL-P-53022 to a thickness of 0.6 to 1.5 mils dry film and painted per MIL-C-46168, color number 383 (green) to a thickness of 1.8 roils minimum dry film (see 4.8.1 and 4.8.2).

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3.6 Identification and marking. The NSG exterior markings shall be lusterless black, color number 37038 of FED-STD-595, and shall conform to MIL-STD-642. Parts and components shall be marked in accordance with MIL-STD-130 (see 4.8.1 and 4.8.2).

3.7 Workmanship. Workmanship shall be of a quality to assure that all new designs procured under the contract shall meet the requirements specified herein. Particular attention shall be paid to marking of parts, plating, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges or similar conditions which could present a safety hazard. Electric wires shall be so located as to prevent possible damage by rubbing on adjacent lines or appendages. No workmanship condition shall exist which may create a safety hazard to operating or maintenance personnel. In addition, all electronic designs must comply with the requirements of MIL-STD-454, Requirement 9 (see 4.8.1 and 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until the conformance of the product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall make available to the Government the documents showing acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity.

4.1.5 Quality assurance requirements. In the conduct of inspection, the contractor shall adhere to Quality Assurance Requirements (QARS) and General Quality Assurance Provisions (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall maintain such certification along with documented test results and performance and analytical data. Such documentation shall be made available upon request by the Government for review.

4.2 Classification of inspections:

a. First article inspection (see 4.3).
   1. Preproduction inspection (PPI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c* Control tests (see 4.5).
d. Comparison tests (see 4.6).

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on the first production NSG and on 2 additional NSG(S), [randomly selected from ten,] which shall be designated as initial production NSG(S). Approval of the first article samples shall not relieve the contractor of the obligation to supply NSG(0) that are fully representative of those inspected as a first article sample. Any change or deviation of production NSG from the first article sample shall be subject to the approval of the contracting officer.

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4.3.1 PPI. When specified (see 6.2), the first NSG produced under a contract shall undergo a preproduction inspection prior to full scale production. The sample NSG shall be subjected to the examinations and tests specified in table VIII, table IX and appendix A. Preproduction approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.3.1.1 In-process inspection. The first production NSG shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures and welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed item inspections.

4.3.1.2.1 Contractor inspection. The first completed production NSG shall be tested and inspected by the contractor, as specified in table VIII and appendix A to determine conformance to the contract and applicable specifications,

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element shall conduct the first production NSG inspection specified in table VIII to the extent that acceptability of the item is verified.

4.3. 1.2.3 Provisional inspection. The Government procuring activity shall perform the first production NSG inspection specified in table VIII to the degree required to justify approval of the item.

4.3.1.3 NSG disposition. After completion of inspection, the first production NSG shall remain at the manufacturing facility, as a production sample, and be the last NSG of its type shipped on the contract. The NSG(S) may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the NSG(S) during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking place after the first production inspection, shall be made to the first production NSG(S) so that the NSG(S) will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on production NSG after Government conditional acceptance of the first production NSG (applicable NSG type) Without written authorization from the Government.

4.3.2 IPT. Two of the first ten production NSGs shall be tested and examined by the Government at a site it shall select. The IPT shell include, as a minimum, the tests and inspections referenced In table VIII. A complete QCI per table VIII shall be performed by the supplier at the place of manufacture before the IPT reliability and maintainability tests are conducted- Upon completion of testing, the supplier shall submit the NSGs (and make available all inspection records and certificates) to the responsible Government inspection element at the supplier's plant for preliminary inspection. The Government at its option may elect to witness and/or participate in the supplier examinations and tests.

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4.3.3 Test failure. Failure of the NSG as a result of any defects of a workmanship or materials nature, during or as a result of the first article inspection, shall be cause for rejection of the NSG. Further, the Government may refuse to continue acceptance of production NSGS until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of the IPT shall be prima facie evidence that all NSGS already accepted prior to completion of the IPT are similarly deficient, unless evidence satisfactory to the contracting officer is furnished by the contractor to verify that they are not similarly deficient- Such defects on all NSGS shall be corrected by the contractor at no cost to the Government.

TABLE VIII. Classification of inspections.

Title	Requirements	Inspection	First article 2/		1/	1/	2/
			PPI	IPT	QCI	Con-trol	Compar-ison
Materials and construction	3.2 thru 3.3.1 and 3.5 thru 3.7	4.8.1	X		X		
Defects (see table IX)	3.2.1 thru 3.3.3, 3.3.6, 3.3.8, 3.3.9, 3.4.7.7, 305 thru 3.7	4.8.2	X		X		
Interface definition	3.3.2	4.8.3			X		
Electrical link	3.3.4	4.8.3			X		
Functional	3.3.5	4.8.3	X			X	
Test equipment interface	3.3.6	4.8.2	X			X	
Weight	3.3.7	4.8.4	X			X	
performance	3.4	4.8.5	X		X		
Preconditioning	3.4. 1	4.8.5.1			X	X	
Power system interface	3.4.2	4.8.5.2	X				
Data input	3.4.3.1	4.8.3			X		
Data link control	3,4.3.1*1	4.8.3			X		
Characteristics of information transferred	3.4.3.1.2	4.8.3				X	
Data rate	3.4.3.1.3	4.8.5.3	X		X		
Character format	3.4.3.1.4	4.8.3			X		
Data character	3.4.3.1.4.1	4.8.3			X		
Numerical data format	3.4.3.1.4.2	4.8.3			X		
Numerical encoding	3.4.3.1.403	4.8.3			X		

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TABLE *VIII.* Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ PPI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Update format	3.4.3.1.4.4	4.8.5.4			x		
Timing	3.4.3.1.5	4.8,5.3	X			x	
Data range	3.4.3.1.6	4.8.3			x		
TSCD to NSG	3.4.3.1.7	4.8.3			x		
Message type	3.4.3.1.7.1	4.8.5.5			x		
Optional data	3.4.3.1.7.2	4.8.5.6	x			x	
End delimiter	3.4.3.1.7.3	4.8.5.6	X			x	
NSG to TSCD	3.4.3.1.8	4.8.3			x		
Start delimiter	3.4.3.1.8.1	4.8.3			x		
Data	3.4,3.1.8.2	4.0.3			x		
End delimiter	3.4.3.1.8.3	4.8.3			x		
NSG modes of operation	3.4.4	4.8.3			x		
NSG modes	3.4.4.1	4.8.5.5			x		
Start up mode	3.4.4.1.1	4.8.5.7			x		
Initialization mode	3.4.4.2	4.8.3			x		
Timing	3.4.4.2.1	4.8.5.7			x		
Waiting	3.4.4.2.2	4.8.5.8	x			x	
Polar data	3.4.4.2.3	4.8.5.9			x		
Directional-gyro mode	3.4.4.3	4.8.5.10			x		
Realign mode	3.4.4.4	4.8.5.6			x		
Polar data	3.4.5.1	4.8.5.3			x		
Examine (X)	3.4.5.2	4.8.5.4			x		
Accuracy	3.4.6	4.8.5.10 and 4.8.5.10.1			x		
Azimuth error	3.4.6.1	4.8.5.10 and 4.8.5.10.1			x		
Vertical angle error	3.4.6.2	4.8.5.10 and 4.8.5. 10.1			x		
Error handling	3.4.7.1	4.8.5.11			x		
Status alert	3.4.7.2	4.8.5.12	x			x	
Status retention	3.4.7.3	4.8.5.13	x			x	
NSG internal failures	3.4.7.4	4.8.5.13	x			x	
Azimuth resolver linearity	3.4.7.5	4.8.5.14	X				
Elevation resolver linearity	3.4.7.6	4.8.5.15	x				

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TABLE VIII. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ PPI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Vehicle motion	3.4.7.7	4.8.2	X			X	
Targeting head motion	3.4.8	4.8.5.16.1 and 4.8,5.16.2	X			X	
Temperature	3.4.9.1	4.8.5.17.1	X				
Humidity	3.4.9.2	4.8.5.17.2	X				
Sand and dust	3.4.9.3	4.8.5.17.3	X				
Salt fog	3.4.9.4	4.8.5.17.4	X				
Water spray	3.4.9.5	4.8.5.17.5	X				
Explosive atmosphere	3.4.9.6	4.8.5.17.6	X				
Electrostatic discharges	3.4.9.7	4.8.5.17.7	X				
Low pressure altitude	3.4.9.8	4.8.5.17.8	X				
Temperature shock	3.4.9.9	4.8.5.17.9	X				
Fungus	3.4.9.10	4.8.5.17.10	X				
Rain	3.4.9.11	4.8.5.17.11	X				
Snow	3.4.9.12	4.8,5.17.12	X				
Icing	3.4.9.13	4.8.5.17.13	X				
Solar radiation	3.4.9.14	4.8.5.17.14	X				
Shock levels	3.4.9.15.1	4,8.5.17.15	X				
Vibration level	3.4.9.15.2	4.8.5.17.16	X				
System Level EMI/EMC	3.4.9.16.1	4.8.5.17.17		X			X
Subsystem Level EMI/EMc	3.4.9.16.2	4.8.5.17.18	X				
Reliability	3.4.10	4.8.18		X			X
Maintainability	3.4.11	4.8.19		X			X

1/ At place of manufacture
2/ At Government proving ground

4.4 QCI. QCI shall be performed in the following sequence and utilizing an approved Final Inspection-Record (FIR).

a. Examination for the defects specified in table VIII and classified in table IX.

b. QCI specified in table VIII.

4.4.1 One hundred percent. Each NSG shall be subjected to the QCI specified in 4.4.

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4.4.2 Acceptable quality level. When authorized by the Government contracting authority, 1 00% QCI (see 4.4.2.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL) on the basis of % defective.

Classification	AQL units
Major	
Minor	150/100 units

4.4.2.1 Sampling

4.4.2.1.1 Lot formation. An inspection lot shall consist of all the NSGs produced in one month submitted at one time for acceptance.

4.4.2.1.2 Sampling for examination. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall examine the first 20 consecutively produced NSG. The examination shall establish that the process average % defective, as defined in MIL-STD-105, is not greater than the specified AQL.

TABLE IX. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 25/100 units	
101	Assembly incomplete (see 3.3 and 3.3.1).	visual
102	Dimensions affecting interchangeability out of tolerance (see 3.3.3 and 3.3.9).	SIE 1/
103	Incorrect or faulty materials, welding defects (see 3.2 and 3.2.1).	Visual
104	Mechanical link, malfunction, damage, improper fit (see 3.3.3 and 3.3.8).	Functional
105	System interface, malfunction, improper fit (see 3.3.2 and 3.3.6).	Functional
106	Identification marking, improper (see 3.6).	Visual
107	Faulty workmanship affecting performance (see 3.4.7.7 and 3.7).	Visual
Minor	AQL 150/100 units	
201	NSG, Improper assembly or installation (see 3.3)*	visual
202	Paint, improper application or color (see 3.5).	visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE - Standard Inspection Equipment.

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4.4.3 Defects. Visual, dimensional, and primary functional examinations shall consist of inspection of the NSG for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and the AQLs specified in Table IX.

4.4.3.1 Unclassified defects. All defects that have no affect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a 30 day period, will be added to the minor defects classification with no increase in AQLs.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in 2 successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.5 Control tests. The NSG shall meet the requirements specified for control tests in table VIII if required by the contract.

4.5.1 Frequency. One of the first 10 NSGS produced, and one per month thereafter, shall be selected for control testing.

4.5.2 Failure. If a NSG fails to pass any control test specified herein, the Government will withhold acceptance inspection and testing on subsequent NSGS until evidence has been provided by the contractor that corrective action has been taken.

4.6 Comparison test. The Government may randomly select NSGS at anytime during the production contract period and subject these samples to the comparison tests specified in table VIII. Tests shall be conducted by the Government at a site it shall select. NSGS selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any NSG to pass any of the specified inspections failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.1, 3.5, 3.6 and 3.7 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor cataloge and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.2.1 through 3.3.3, 3.3.6, 3.3.8, 3.3.9, 3.4.7.7, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in table IX. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Demonstration of features. Conformance to 3.3.2, 3.3.4, 3.3.5, 3.4.3.1 through 3.4.3.1.2, 3.4.3.1.4 through 3.4.3.1.4.3, 3.4.3.1.6, 3.4.3.1.7, 3.4.3.1.8.1 through 3.4.4, and 3.4.4.2 shall be demonstrated and/or exercised and shown to be present and/or functional in a qualitative marine r. Such demonstrations may be combined with quantitative tests or inspections when such tests or inspections clearly demonstrate or exercise the feature and/or function.

4.8.4 Weight. Conformance to 3.3.7 shall be determined by weighing the NSG on a scale.

4.8.5 Performance. To determine conformance to 3.4, the NSG shall -be simulated to IM and while in that mode the unit shall be moved through an arc of not less than 60 degrees in roll and pitch at rates greater than 20 degrees per second. Subsequent proper operation shall not indicate occurrence of any physical damage or degradation to the NSG or its components.

4.8.5.1 Preconditioning. To determine conformance to 3.4.1, each unit shall be preconditioned as specified in 4.8.5.1.1 through 4.8.5.1.3.

4.8.5.1.1 Bum-in, As part of, or prior to, QCI at normal ambient environment, each unit shall be subjected to thermal stress cycling designed to induce infant failure at the manufacturer's facility. The thermal stress temperatures shall be as follows:

Cold Soak Temperature: -40°F
Hot Soak Temperature: +160°F
Operating Temperature: -25°F to +145°F

The unit shall be placed in an appropriate burn-in chamber and shall be temperature cycled as shown in figure 6. The temperature cycle shall be repeated until 3 consecutive failure-free transitions have been completed. Functional/electrical tests shall be conducted during the temperature extremes.

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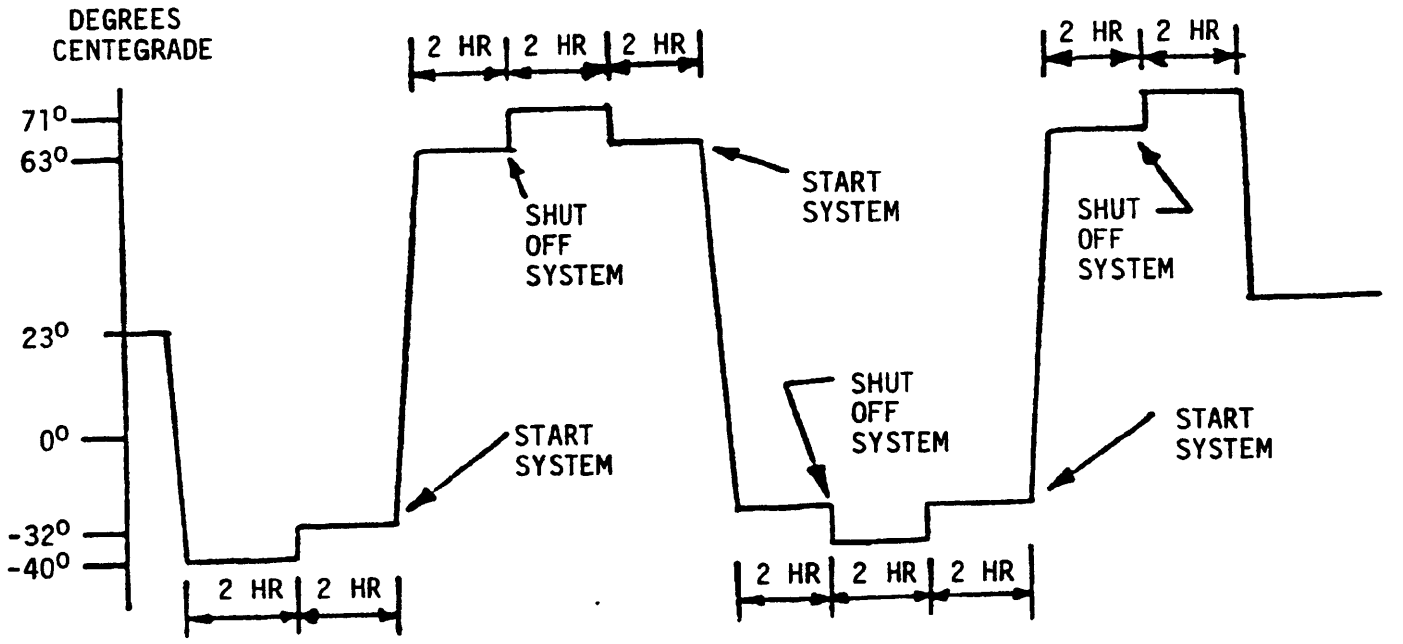


FIGURE 6. FISTV/NSG temperature cycle.

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4.8.5.1.2 Shake out. The equipment, while operating and supported at its mounting points, shall be subjected to random vibration having a root mean square (RMS) amplitude of 2.9g, RMS ± 3.0%, and a relative acceleration spectral density as shown in figure 7 for a period of 30 minutes. During the shake out the vibration input shall be vertical with respect to earth. This test may be run concurrently with the operation test of paragraph 4.8.5.1.1. The unit shall be in the D-G mode during this shake out test and shall be monitored in accordance with paragraph 4.8.5.1.3.

4.8.5.1.3 Preconditioned monitoring. The NSG shall be monitored during preconditioning in such a way that performance anomalies induced by preconditioning will be identified.

4.8.5.2 Power system interface. To determine conformance to 3.4.2, the power system interface shall be tested as specified in 4.8.5.2.1 through 4.8.5.2.5.

4.8.5.2.1 Input power. The input power shall be measured during each mode and shall not exceed the specified requirements.

4.8.5.2.2 Voltage. The input voltage shall be set to the specified lower operating extreme, and paragraph 4.8.5.10b shall be executed.

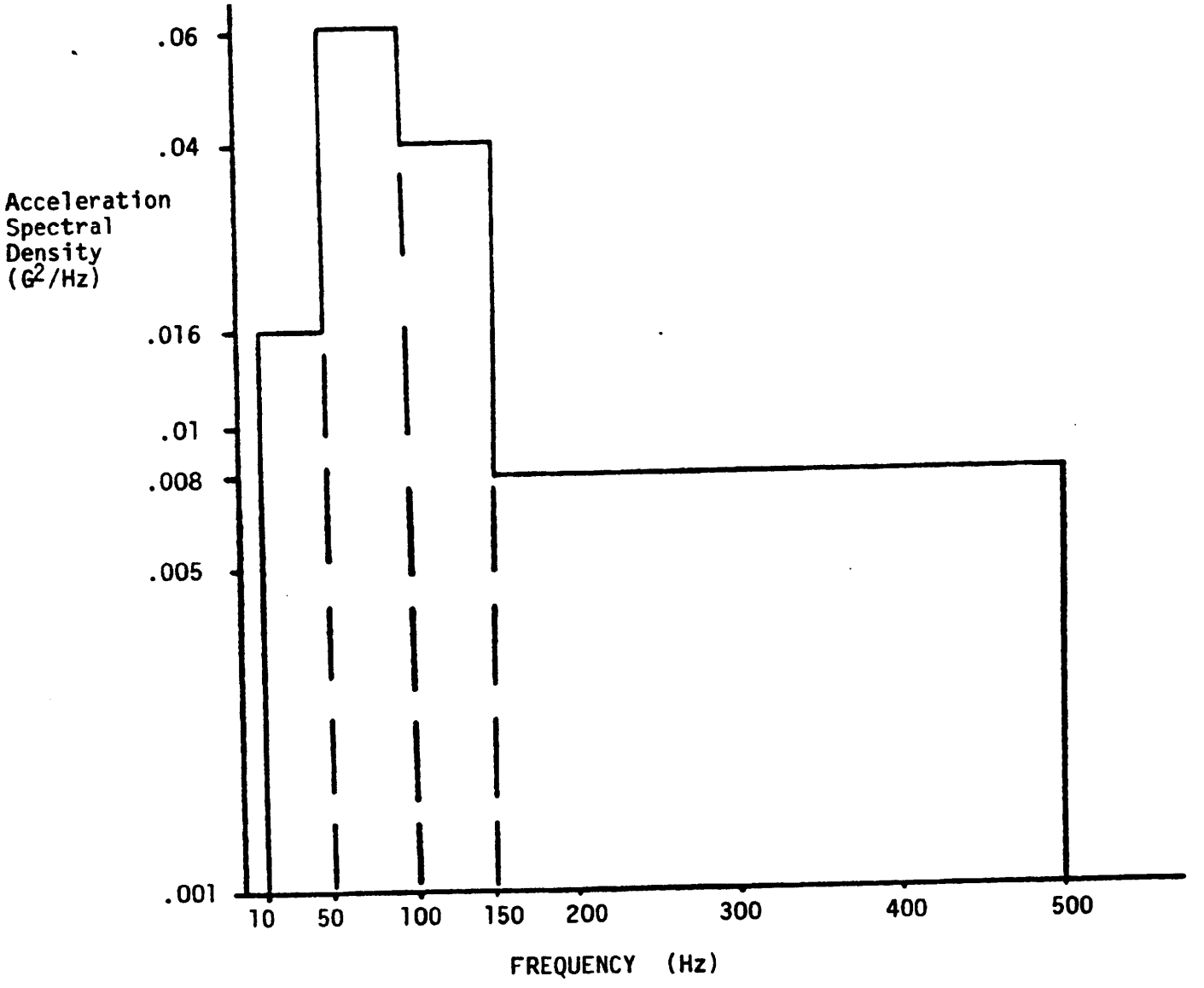
4.8.5.2.3 Supply anomaly I. The NSG shall be subjected to supply variations characteristic of the worst case limits of MIL-STD-1275 while operating in either D-GM or IM. The NSG shall not be damaged. Verification of no damage shall be by execution of paragraph 4.8.

4.8.5.2.4 Supply anomaly II. The equipment, while in each of its modes, shall be subjected to supply variations defined below. Whenever the supply voltage is within the specified operating range, the NSG shall perform as specified. The NSG shall be interrogated for mode and status every 10 seconds during this test; in no event shall the NSG transmit erroneous data. The supply variations are as follows: using a voltage rate of change of not less than 1 volt per second nor more than 3 volts per second, start at 24 V dc, reduce to 8 V dc, dwell for 10 seconds, increase to 19.5 V dc, dwell for 10 seconds, reduce to 12 V dc, dwell for 10 seconds, increase to 20.5 V dc, dwell for 30 seconds.

4.8.5.2.5 Loss of power. The unit shall be sequenced to each of its modes. Once during each mode, the power shall be interrupted for a period of not less then .5 seconds nor more than 1.5 seconds. Subsequent proper operation shall not indicate occurrence of any physical damage or degradation to the NSG or its components.

4.8.5.3 Data rate. To determine conformance to 3.4.3.1.3, 3.4.3.1.5 and 3.4.5.1, the NSG shall be stimulated with each of the message types while the NSG is In each of its modes. The response timing and sequence shall be measured as shown in figure 3 and shall be as specified. Alternate verification by means of analysis is permissible.

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NOTE: Due to limitations of vibration equipment, the applied acceleration may vary up to ±25% of nominal at a given frequency with breakpoints of up to 3 Hz on either side of the specified breakpoint frequency. The tolerance of the RMS acceleration shall be as specified elsewhere herein.

FIGURE 7. Acceleration spectral density versus frequency.

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4.8.5.4 Update format. To determine conformance to 3.4.3.1.4.4 and 3.4.5.2, while the NSG is in each mode, the NSG shall be stimulated with each of the message types and the NSG response obtained shall be in accordance with table IV and as specified elsewhere herein. NSG accuracy shall be as specified for each response to request for polar data.

4.8.5.5 Message type. To determine conformance to 3.4.3.1.7.1 and 3.4.4.1, while the NSG Is in each of its modes, the NSG shall be stimulated with a properly formatted command except that the message type shall be erroneous. The NSG shall respond as specified.

4.8.5.6 Optional data. To determine conformance to 3.4.3.1.7.2, 3.4.3.1.7.3 and 3.4.4.4, while in each of its modes, the NSG shall be stimulated with an incorrectly formatted, but otherwise legitimate message type, and shall respond as specified.

4.8.5.7 Start up mode. To determine conformance to 3.4.4.1.1 and 3.4.4.2.1, the NSG shall be sequenced through each of its modes and each mode shall be timed and shall be as specified. Timing shall be measured as the time between transition points.

4.8.5.8 Waiting. To determine conformance to 3.4.4.2.2, the NSG shall be energized and allowed to sequence without further stimulus; the NSG shall sequence as specified and shall WAIT. The NSG shall be allowed to WAIT 10 minutes at the conclusion of which time an UPDATE transmission shall be used to stimulate the NSG whereupon it shall then proceed as specified.

4.8.5.9 Polar data. To determine conformance to 3.4.4.2.3, the NSG shall be energized and allowed to sequence normally and shall be provided with proper UPDATE transmission during the first two minutes from application of power; the NSG shall be observed to sequence through START UP, INITIALIZE and into D-GM without further stimulation.

4.8.5.10 Accuracy. To determine conformance to 3.4.4.3, 3.4.6, 3.4.6.1 and 3.4.6.2, the NSG accuracy test shall be as specified below:

a. The NSG shall be positioned to 0 roils in azimuth, elevation and roll; paragraph 4.8.5.10.1 shall be executed.

b. The NSG shall be positioned to various known orientations within its specified operation range. The unit shall be stimulated to respond with its present angular coordinates. Data obtained shall be compared against the known coordinates and the error shall not exceed the specified accuracy requirement, nor shall it indicate a trend which would result in exceeding the required accuracy.

c. Subsequent to the completion of step a above, the NSG shall be positioned to 600 $\pm$ 15 roils elevation, 300 + 15 roils roll, and any azimuth angle. The unit shall be stimulated to respond with its present angular coordinates. After a minimum of 15

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minutes, the unit shall again be stimulated to respond with its angular coordinates. The latter responses shall be compared to the former response and the difference shall not indicate a trend which would result in exceeding the required accuracy.

4.8.5.10.1 Data accuracy. The NSG shall be properly inputted in accordance with local latitudinal and longitudinal coordinates while oriented as specified and shall be initialized and sequenced to the D-G mode. The NSG shall be stimulated by the (P) Request for Polar coordinates command or appropriate examine command. Data obtained shall be compared against the specified coordinates and the error shall not exceed the specified accuracy requirements nor shall it indicate a trend which would result in exceeding the required accuracy.

4.8.5.11 Error handling. To determine conformance to 3.4.7.1, while in each mode, the NSG shall be stimulated by each of the legitimate message types, correctly formatted and protocalled except that the parity shall be in error. The NSG response shall be as specified.

4.8.5.12 Status alert. To determine conformance to 3.4.7.2, while the NSG is in each of its modes, the Status Alert Line shall be observed and its behavior shall be as specified at each transition point whereupon a Request for Polar Data (P) shall be used to stimulate the NSG. The Status Alert Line shall not change and the NSG shall then be stimulated with a Status Request (XO02) whereupon the NSG shall respond as specified.

4.8.5.13 NSG status retention and internal failure. To determine conformance to 3.4.7.3 and 3.4.7.4, each failure type identified in table VII shall be simulated or induced (non-destructively) while the NSG is in an appropriate mode. The Status Alert Line shall be observed and the NSG shall be stimulated with the Status Request (XO02) message. The NSG shall respond as specified. The unit shall remain energized and a simulated failure shall be removed (test 1 type). The NSG shall respond as specified when appropriately stimulated (the NSG may be initialized if required). Alternate verification by means of analysis is permissible.

4.8.5.14 Azimuth resolver linearity. To determine conformance to 3.4.7.5, the system shall be mounted on a Goertz (or equivalent) table aligned north. The system shall be turned on as specified in appendix A. After the system has reached the D-GM, rotate the table in 15° increments for 360°. Polar data shall be requested and recorded at each position.

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Error (1A) _____ roils < ± 2 mils

$$1\Delta = \sqrt{\frac{(Xi - X)^2}{n-1}} \quad \text{for} \quad i=1 \quad \text{ton}$$

where n = number of samples

x = mean of the samples

Xi = data point

4.8.5.15 Elevation resolver linearity. To determine conformance to . 3.4.7.6, the system shall be mounted on a tilt table in a level position. The system shall be turned on as specified in appendix A. After the system has reached the X)-GM, elevate the system in 15° Increments from -45° to +60°. Polar data shall be requested and recorded at each position.

Error (1A) = _____ mils < ± 3 roils

4.8.5.16 Targeting head motion. To determine conformance to 3.4.8, the NSG motion shall be tested as follows:

4.8.5.16.1 Combined motion. The NSG shall be placed in a gimballed fixture and, while in D-GM, it shall be moved in elevation (pitch) and roll along a linear locus connecting the extremes of travel necessitated by the specified combined vehicle and launcher head angular travel. Accuracy and performance shall meet specified requirements during and after this test.

4.8.5.16.2 Rotational motion. The NSG shall be placed on a rotation device whose axis of rotation is off local vertical by 20°. The NSG shall be sequenced to D-GM and shall be oriented to a known position at which time the NSG shall be stimulated to transmit azimuth and elevation data. The NSG shall then be rotated on the device at not less than 200° per second in a positive direction for not less than 18 seconds at the conclusion of which the unit shall be reoriented to the known position whereupon the NSG shall be stimulated to transmit azimuth and elevation data. The comparison of the before and after data shall indicate compliance with performance requirements.

4.8.5.17 Environmental conditions.

4.8.5.17.1 Temperature test. To determine conformance to 3.4.9.1, the NSG shall be tested as follows:

a. Operating temperature

1. Low temperature. The NSG shall be tested in accordance with MIL-STD-81OC, method 502.1, procedure I at -25°F. Prior to testing and at low operating temperature, the NSG shall be tested in accordance with 4.8.5.4.

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2. High temperature. The NSG shall be tested in accordance with MIL-STD-810C, method 501.1, procedure I at +145°F. Prior to testing and at high operating temperature, the NSG shall be tested in accordance with 4.8.5.4.

b. Nonoperating (storage) temperature

1. Low temperature. The NSG shall be tested in accordance with MIL-STD-810C, method 502.1, procedure I at -40°F for 24 hours duration.

2. High temperature. The NSG shall be tested in accordance with MIL-STD-810C, method 501.1, procedure I at +160°F for 24 hours duration.

4.8.5.17.2 Humidity. To determine conformance to 3.4.9.2, the NSG shall be operated during the humidity test in accordance with MIL-STD 810, method 507, procedure I, with the exception that the high temperature shall be +145°F and the low temperature shall be -25°F. The test shall be repeated with a low relative humidity of 5 percent at temperatures of +70°F and +145°F.

4.8.5.17.3 Sand and dust. To determine conformance to 3.4.9.3, the NSG shall be operated during both sand and dust tests in accordance with MIL-STD-810, method 510, procedure I, with the exception that the sand particle size shall be 0.0lmm to 1.0mm in diameter, with wind greater than 30 knots; and the dust particle size shall be 0.0001mm to 0.0lmm diameter, and density of 6 x 10^{-9} gm/cm^3 with a wind of 35 knots.

4.8.5.17.4 Salt fog. To determine conformance to 3.4.9.4, the NSG shall be operated during a salt fog test in accordance with MIL-STD-810, method 509, procedure I.

4.8.5.17.5 Water spray. To determine conformance to 3.4.9.5, the NSG electrical and electronic equipment shall be tested to show that they are spray tight as defined in MIL-STD-108.

4.8.5.17.6 Explosive atmosphere. To determine conformance to 3.4.9.6, the NSG shall be tested to the explosive atmosphere test requirements of MIL-STD-810, method 511, procedure I.

4.8.5.17.7 Electrostatic discharges. To determine conformance 3.4.9.7, the NSG shall be tested to the electrostatic discharge requirements.

4.8.5.17.8 Pressure test (altitude). To determine conformance to 3.4.9.8, the NSG shall be tested in accordance with MIL-STD-810, method 500, procedure 1 as follows:

a. Operating. The air pressure shall be equivalent to an altitude of 10 000 feet.

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b. Nonoperating. The air pressure shall be equivalent to an altitude of 40 000 feet.

4.8.5.17.9 Temperature shock. To determine conformance to 3.4.9.9, the NSG shall be tested in accordance with MIL-STD-81O, method 503, procedure I, except that the low temperature shall be -40°F.

4.8.5.17.10 Fungus. To determine conformance to 3.4.9.10, the NSG shall be tested at 84°F and 95% relative humidity with fungi as specified in MIL-StD-81O, method 508, for a period of 90 days.

4.8.5.17.11 Rain. To determine conformance to 3.4.9.11, a rain test shall be performedin accordance with method 506, procedure I of MIL-STD-81O, with the exception that duration and rainfall be as specified in 3.4.9.11.

4.8.5.17.12 Snow. To determine conformance to 3.4.9.12, the NSG shall be operated during the following snow tests:

a. Falling snow. A falling snow test shall be performed using a suitable environmental chamber to show that the NSG operates in falling snow with crystal size range of 0.05 to 20mm diameter (median diameter 2.0 to 5.Omm).

b. Blowing snow. A blowing snow test shall be performed using a suitable environmental chamber to show that the NSG operates in blowing snow with crystal size range of 0.02 to 0.4mm . diameter and wind velocity above 10 knots.

4.8.5.17.13 Icing (nonoperating). To determine conformance to 3.4.9.13, the NSG shall be tested as specified below:

a. Frost. A frost test shall be performed using a suitable environmental chamber to show that the NSG is not damaged by deposition of surface frost of ice crystals averaging 1/16 inch to 1/4 inch in diameter.

b. Rime icing. A rime ice test shall be performed using a suitable environmental chamber to show that the NSG is not damaged by deposition of rime ice crystals averaging 0.056mm in diameter.

C. Clear glaze. A clear glaze test shall be performed using a suitable environmental chamber to show that the NSG is not damaged by deposition of 2.0 inch clear glaze ice with a specific gravity of 0.85.

4.8.5.17.14 Solar radiation. To determine conformance to 3.4.9.14, a solar radiation test shall be performed in accordance with MIL-STD-810), method 505, procedure I.

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4.8.5.17.15 Shock. To determine conformance to 3.4.9.15.1, the test specimen shall be tested to the requirements of MIL-STD-810, method 516, procedure 1 for ground equipment. Either figure 516 (11 millisecond sawtooth pulse with 40g peak) or figure 516.2-1 (11 millisecond half sine wave pulse with 30g peak) may be used with the sawtooth shock preferred*

4.8.5.17.16 Vibration. To determine conformance to 3.4.9.15.2, the NSG shall be tested in accordance with MIL-STD-810, method 514, procedure VIII, curve W of figure 514.2-6 with a sinusoidal cycling time of 2.5 hours and a maximum of 3 hours. Sweep time is 15 minutes from 5 to 500 to 5 Hz. Curve W imposes 1.0 inch double amplitude vibration for 5 to 5.5 Hz, 1.5g from 5.5 to 30 Hz, 0.033 inch double amplitude from 30 to SO Hz, and 4.2g from 50 to 500 Hz.

4.8.5.17.17 System Level EMI/EMC. To determine conformance to 3.4.9.16.1, the NSG shall be tested on the vehicle in accordance with MIL-STD-462, RE05 and CE07 test procedures.

4.8.5.17.18 Subsystem Level EMI/EMC. To determine conformance to 3.4.9.16.2, the NSG shall be tested off the vehicle in accordance with the applicable MIL-STD-462 test procedures.

4.8.18 Reliability. To verify conformance to 3.4.10, a verification of reliability shall be performed during IPT. The requirements for NSG reliability shall be as stated in 3.4.10.

4.8.19 Maintainability. To verify conformance to 3.4.11, a maintenance ratio (MR) equal to or less than that specified in 3.4.11 must be demonstrated during IPT testing. MR equals the calendar maintenance time divided by the system operating time.

5* PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The NSG is intended to provide azimuth and elevation heading information referenced to grid north and local vertical for the FISTV system.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. NSG Drawing 12306660 required.
If first article samples are not required (see 3.1).
d. If painting shall be other than as specified (see 3.5).
e. If system level EMI/EMC shall be other than specified (see 3.4.9.16.2).

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f. If responsibility for inspection shall be other than as specified (see 4.l).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
h. If preproduction inspection is required (see 4.3.1).
i. Selection of applicable level of preservation, packaging and packing (see 5.1).

6.3 Process average. Sampling may be initiated if the process average value for the first 20 NSGs inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{process average} = \frac{\text{number of defects}}{\text{number of NSGs inspected}} \times 100$$

If the computed average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for 20 consecutive NSGs is less than the specified AQL.

6.4 Definitions.

6.4.1 Recovered materials. 'Recovered materials' means materials that have been collected or recovered form solid waste (see 6.4.2) to become a source of new materials, as distinguished from virgin raw materials.

6.4.2 Solid waste. For definition of 'solid waste', refer to section 23.402 of the Federal Acquisition Regulations.

6.5 Subject (key word) listing.

FISTV, M981, North Seeking, Gyrocompass
Gyrocompass Assembly, North Seeking

6.6 Supersession data. This military specification supersedes M981PD-G-62487, dated 24 January 1986.

Custodian:
Army - AT

Preparing activity:
Army - AT

(project 1290-A395)

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APPENDIX A

Accuracy Test Course

10. SCOPE.

10.1 Scope. This appendix provides instructions for the accuracy test course.

20. REFERENCED DOCUMENT. Not applicable.

30. Requirements.

30.1 ACCURACY TEST COURSE.

30.1.1 Power-on sequence.

a. Turn on 24 V NSG system power and enter the appropriate Northing and Easting data an an update command.

Example: for Teterboro, NJ location:

U57850004523000<

Easting 578500
Northing 4523000
0 North hemisphere
1 South hemisphere

b. After the power has been turned on, monitor "Status Alert" signal, pin z (relative to pin as i.e., differential). A logic level "1" occurs when a mode changes and/or BITE failure occurs . Upon this signal, transmit an examine command XO02< to the NSG. This will reset z and reveal whether a BITE failure has ocurred.

c. During system initialization, the response to the XO02 examine commands shall follow the following sequence:

1 >0000< Start Up (≤ 3 minutes in mode)
2 >0100< Coarse Align
3 >0200< Fine Align
4 >0300< D-GM (≤ 10 minutes from power "on")

During a realign, the response to the XO02< examine command shall be >0400<. Start of realign to start of D-GM should be ≤ 5 minutes.

If the last two digits of the response are not zero, a system failure has been detected by the BITE circuitry. The test sequence will stop until the failure is identified and corrected.

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d. Once the system is in the D-GM, request polar information by transmitting a P< to the NSG. The system will respond with azimuth and elevation values. If the system response is a retransmit the P<. The azimuth and elevation values should be within ± 4 mils and ± 3 mils of the actual.

30.1.2 Realign. A realign is initiated by transmitting an R<. Time and examine the NSG with the "XO02" command when the STATUS alert signal is set. Request polar information as per 30.1.ld.

30.2 System test.

a. Power usage. Turn on the power to the NSG and record the supply voltage and peak current within 5 seconds. Respond to the system as per paragraphs 30.1.la and 30.1.lb. Measure the input voltage and record it. Once the NSG has reached the D-GM, measure and record the supply current. The start power should be <200 watts and the run power <120 watts.

b. Status alert - mode timing.

c. Accuracy test (see paragraph 3.4.6). Mount the NSG on a position table with the system pointing north. The table is required to position the unit under test (UUT) level and North, South, East and West to within 0.5 roils of the position known to 0.1 mil.

d. Nominal voltage.

(1) Turn on the system as per 30.1.la and 30.1.lb. Record the actual position. When the system reaches the D-GM, request polar data by transmitting a P<. Record the polar data response.

(2) Start a drift test by transmitting D< to the NSG. Allow the system to drift for 6 minutes, recording the response to the high resolution azimuth command XOF2< and elevation command X122< command every minute.

(3) perform 5 realigns as per 30.1.2 and record polar data response.

(4) Rotate the system 1600 roils to the east and record the new position in the sample data set below. Perform a reinitialize by transmitting an I<. Record the polar data response when the system reaches the D-GM.

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(5) Start a drift test by transmitting D<. Allow the system to drift for 6 minutes, recording the response to the high resolution azimuth command XOF2< and elevation command X122< command every minute.

(6) perform 5 realigns as per paragprah 30.1.2 and record polar information.

e. Voltage Sensitivity

(1) Set supply voltage to 20 V dc and record voltage. Position the system South and repeat paragraph 30.2d(4).

(2) Set supply voltage to 30 V dc and record voltage. Position the UUT West and repeat paragraph 30.2d(4). Reset power supply to 24 V dc.

f. Elevation range. With the system mounted on an elevation position table and in the D-GM, position the UUT to -53° and +68° . Request and record polar data at the extreme positions.

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